JT808 Proactive Safety Additional Information Protocol

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Version History

Version	Date	Modifier	Change Description
1.0.0	2026-04-22	Zhuojie Cai	Initial version
1.0.1	2026-05-23	Zhuojie Cai	Added complete frame examples for all alarm protocols (0x64, 0x65, 0x66, 0x67, 0x70, 0xA1); optimized Appendix A alarm attachment upload process structure
1.0.2	2026-06-11	Zhuojie Cai	Added IPC alarm/event type fire and smoke detection (0xE0)

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Scope

This protocol specifies the data format of location additional information items in the Active Safety Intelligent Prevention and Control System.

Normative References

- Road Transport Vehicle Active Safety Intelligent Prevention and Control System Technical Specification Part 3: Communication Protocol

This section summarizes the complete definitions of alarm types, distinguishing between **Active Safety Standard Definitions** and **[Manufacturer Custom] Definitions**.

For alarm attachment upload process, see **Appendix A Alarm Attachment Upload Process**.

Note:

- **Standard Definitions**: Source from Road Transport Vehicle Active Safety Intelligent Prevention and Control System Technical Specification
- **[Manufacturer Custom]**: internal custom extension types, source from historical drafts or manufacturer customizations

Manufacturer Custom Extension Recommendation: For future new alarm/event types, it is recommended to use the range **0xE0~0xFF**, and allocate as far back as possible (e.g., 0xE0, 0xE1).
Reason: Custom IDs should not be too close to standard IDs. When the protocol is updated in future versions, it will accumulate forward from the existing base. Maintaining a skipping allocation avoids conflicts.

> **Remark:** For other extension protocols or if you cannot find the required protocol definition, please contact **Software Technical Support Team**.

1 Proactive Safety Additional Information Item Format (TLV Structure)

Additional information items adopt the TLV (Tag-Length-Value) structure:

Additional Information ID (1 Byte) + Additional Information Length (1 Byte) + Additional Information Data (N Byte)

Field	Data Type	Description and Requirements
Additional Information ID	BYTE	1~255
Additional Information Length	BYTE	—
Additional Information	—	Additional information definitions see Table 2-1

2 Additional Information Definitions

Table 2-1 Additional Information Definitions

Additional Information ID	Additional Information Length	Description and Requirements

0x64	47	Driving assistance function alarm information, see Table 3-1
0x65	52	Driver behavior monitoring function alarm information, see Table 4-1
0x66	41+8n	Tire status monitoring alarm information, see Table 5-1 (n is tire count)
0x67	41	Blind spot monitoring alarm information, see Table 6-1
0x70	47	Aggressive driving alarm information, see Table 7-1
0xA1	72	IPC alarm events, see Table 8-1

3 Driving Assistance Function Alarm Information (0x64)

Table 3-1 Driving Assistance Function Alarm Information Data Format

Offset	Field	Data Type	Description and Requirements
0	Alarm ID	DWORD	Cumulative increment from 0 in alarm order, regardless of alarm type
4	Flag Status	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag

5	Alarm/Event Type	BYTE	**Standard Definitions**: 0x01: Forward collision warning 0x02: Lane departure warning 0x03: Too close to vehicle ahead warning 0x04: Pedestrian collision warning 0x05: Frequent lane change warning 0x06: Road sign exceedance warning 0x07: Obstacle warning 0x08: Driving assistance function failure warning **[Manufacturer Custom] Definitions**: 0x09: Vibration alarm 0x0A: Overspeed 0x0B: Startup (ACC startup event) 0x0C: Shutdown (ACC shutdown event) 0x0D: GSensor collision alarm 0x12: Vehicle ahead moving away 0xA1: Harsh acceleration

			0xA2: Harsh braking 0xA3: Harsh cornering 0xA4: Pedestrian attention 0xAC: Fuel theft 0xAD: Fuel refueling 0xAF: Rollover alarm event
6	Alarm Level	BYTE	0x01: Level 1 alarm 0x02: Level 2 alarm
7	Preceding Vehicle Speed	BYTE	Unit: km/h, range 0~250 Only valid when alarm type is 0x01 and 0x02
8	Preceding Vehicle/Pedestrian Distance	BYTE	Unit: 100ms, range 0~100 Only valid when alarm type is 0x01, 0x02, and 0x04
9	Deviation Type	BYTE	0x01: Left deviation 0x02: Right deviation Only valid when alarm type is 0x02
10	Road Sign Recognition Type	BYTE	0x01: Speed limit sign 0x02: Height limit sign 0x03: Weight limit sign Only valid when alarm type is 0x06

11	Road Sign Recognition Data	BYTE	Recognized road sign data
12	Vehicle Speed	BYTE	Unit: km/h, range 0~250
13	Elevation	WORD	Altitude in meters (m)
15	Latitude	DWORD	Latitude value in degrees multiplied by 10 to the 6th power, accurate to 1/1,000,000 degree
19	Longitude	DWORD	Longitude value in degrees multiplied by 10 to the 6th power, accurate to 1/1,000,000 degree
23	Date Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)
29	Vehicle Status	WORD	See Table 3-2
31	Alarm ID Number	BYTE[16]	See Appendix A.4

Table 3-2 Vehicle Status Bit Definitions

Bit	Definition	Description
0	ACC Status	0: Off, 1: On
1	Left Turn Signal Status	0: Off, 1: On
2	Right Turn Signal Status	0: Off, 1: On
3	Wiper Status	0: Off, 1: On
4	Brake Status	0: Not braking, 1: Braking
5	Card Insertion Status	0: Card not inserted, 1: Card inserted

6~9	Reserved	—
10	Positioning Status	0: Not positioned, 1: Positioned
11~15	Reserved	—

Example Data (47 Bytes):

Scenario: Forward collision warning (0x01), start flag, level 1 alarm

Field	HEX Value	Description
Alarm ID	`00 00 00 01`	Sequence 1
Flag Status	`01`	Start flag
Alarm/Event Type	`01`	Forward collision warning
Alarm Level	`01`	Level 1 alarm
Preceding Vehicle Speed	`64`	100 km/h
Preceding Vehicle/Pedestrian Distance	`32`	50 × 100ms = 5 seconds
Deviation Type	`01`	Left deviation (reserved)
Road Sign Recognition Type	`01`	Speed limit sign (reserved)
Road Sign Recognition Data	`64`	Recognition data (reserved)
Vehicle Speed	`64`	100 km/h
Elevation	`03 E8`	1000 meters
Latitude	`0F 42 3D FF`	32,413,583 × 10^{-6}° ≈ 32.413583°
Longitude	`0D ED BA 52`	233,862,360 × 10^{-6}° ≈ 233.862360°
Date Time	`32 04 16 08 1E 2D`	2026-04-22 08:30:45

Vehicle Status	`00 00`	No status bits
Alarm ID Number	`00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08`	—

Complete HEX:

64 2F 01 01 01 01 64 32 01 01 64 64 03 E8 0F 42 3D FF 0D ED BA 52 32 04 16 08 1E 2D 00 00 00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08

Parsing: Additional Information ID=64 (Driving Assistance), Additional Information Length=2F (47 Bytes), followed by 47 Bytes of data content

Complete Frame Example (Uplink): 7E 02 00 00 4D 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 64 EC 15 06 BE 20 80 00 00 00 25 80 05 A2 60 51 31 51 15 96 64 2F 00 00 00 01 01 01 01 96 64 01 01 64 64 00 64 01 EE 97 8F 07 14 7D 01 E0 26 05 13 14 36 56 00 00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08 C4 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 0D 00 01 02 00 00 00 7E

4 Driver Behavior Monitoring Function Alarm Information (0x65)

Table 4-1 Driver Behavior Monitoring Function Alarm Information Data Format

Offset	Field	Data Type	Description and Requirements
0	Alarm ID	DWORD	Cumulative increment from 0 in alarm order, regardless of alarm type
4	Flag Status	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag

5	Alarm/Event Type	BYTE	**Standard Definitions**: 0x01: Fatigue driving warning 0x02: Handheld phone call warning 0x03: Smoking warning 0x04: Long time not looking ahead warning 0x05: Driver not detected warning 0x06: Both hands off steering wheel warning 0x07: Driver behavior monitoring function failure warning 0x08~0x0F: User defined (reserved) 0x10: Auto capture event 0x11: Driver change event **[Manufacturer Custom] Definitions**: 0x0A: Seatbelt not fastened 0x12: Infrared blocking sunglasses failure reminder 0x13: Device occlusion failure reminder 0x14: Face statistics

			0x15: Face cropping 0x16: Card not swiped warning 0x17: Driver identity recognition event 0x1F: Mask not worn warning 0xA3: Eye closed 0xA4: Yawning
6	Alarm Level	BYTE	0x01: Level 1 alarm 0x02: Level 2 alarm
7	Fatigue Level	BYTE	Range 1~10, higher value indicates more severe fatigue Only valid when alarm type is 0x01
8	Reserved	BYTE[4]	Reserved
12	Vehicle Speed	BYTE	Unit: km/h, range 0~250
13	Elevation	WORD	Altitude in meters (m)
15	Latitude	DWORD	Latitude value in degrees multiplied by 10 to the 6th power
19	Longitude	DWORD	Longitude value in degrees multiplied by 10 to the 6th power
23	Date Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)
29	Vehicle Status	WORD	See Table 4-2

31	Alarm ID Number	BYTE[16]	See Appendix A.4

Table 4-2 Vehicle Status Bit Definitions

Bit	Definition	Description
0	ACC Status	0: Off, 1: On
1	Left Turn Signal Status	0: Off, 1: On
2	Right Turn Signal Status	0: Off, 1: On
3	Wiper Status	0: Off, 1: On
4	Brake Status	0: Not braking, 1: Braking
5	Card Insertion Status	0: Card not inserted, 1: Card inserted
6~9	Reserved	—
10	Positioning Status	0: Not positioned, 1: Positioned
11~15	Reserved	—

Example Data (47 Bytes):

Scenario: Fatigue driving warning (0x01), start flag, level 1 alarm, fatigue level 5

Field	HEX Value	Description
Alarm ID	`00 00 00 02`	Sequence 2
Flag Status	`01`	Start flag
Alarm/Event Type	`01`	Fatigue driving warning
Alarm Level	`01`	Level 1 alarm
Fatigue Level	`05`	Level value 5
Reserved	`00 00 00 00`	Reserved
Vehicle Speed	`32`	50 km/h

Elevation	`01 F4`	500 meters
Latitude	`0F 42 3D FF`	32,413,583 × 10^{-6}° ≈ 32.413583°
Longitude	`0D ED BA 52`	233,862,360 × 10^{-6}° ≈ 233.862360°
Date Time	`32 04 16 08 1E 2D`	2026-04-22 08:30:45
Vehicle Status	`00 00`	No status bits
Alarm ID Number	`00 00 00 02 00 00 00 02 00 02 03 04 05 06 07 08`	—

Complete HEX:

65 2F 02 01 01 01 05 00 00 00 00 32 01 F4 0F 42 3D FF 0D ED BA 52 32 04 16 08 1E 2D 00 00 00 00 00 02 00 00 00 02 00 02 03 04 05 06 07 08

Complete Frame Example (Uplink): 7E 02 00 00 4D 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 65 EC 15 06 BE 20 80 00 00 00 25 80 05 A2 60 51 31 52 15 16 52 F0 00 00 00 01 00 01 01 05 00 00 00 00 32 00 64 01 EE 97 8F 07 14 7D 01 E0 26 05 13 14 36 56 00 00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08 39 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 0E 00 01 02 00 00 03 7E

5 Tire Status Monitoring Alarm Information (0x66)

Table 5-1 Tire Status Monitoring Alarm Information Data Format

Offset	Field	Data Type	Description and Requirements
0	Alarm ID	DWORD	Cumulative increment from 0 in alarm order, regardless of alarm type

4	Flag Status	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag
5	Vehicle Speed	BYTE	Unit: km/h, range 0~250
6	Elevation	WORD	Altitude in meters (m)
8	Latitude	DWORD	Latitude value in degrees multiplied by 10 to the 6th power
12	Longitude	DWORD	Longitude value in degrees multiplied by 10 to the 6th power
16	Date Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)
22	Vehicle Status	WORD	See Table 5-2
24	Alarm ID Number	BYTE[16]	Alarm ID number definition, see Appendix A.4
40	Total Alarm/Event List Count	BYTE	Number of alarm/event information items
41	Alarm/Event Information List	—	List format

Table 5-2 Vehicle Status Bit Definitions

Bit	Definition	Description
0	ACC Status	0: Off, 1: On
1	Left Turn Signal Status	0: Off, 1: On

2	Right Turn Signal Status	0: Off, 1: On
3	Wiper Status	0: Off, 1: On
4	Brake Status	0: Not braking, 1: Braking
5	Card Insertion Status	0: Card not inserted, 1: Card inserted
6~9	Reserved	—
10	Positioning Status	0: Not positioned, 1: Positioned
11~15	Reserved	—

Table 5-1 Tire Status Monitoring Function Alarm Information List Format

Offset	Field	Data Type	Description
0	Tire Alarm Position	BYTE	Tire position number
1	Alarm/Event Type	WORD	bit0: Tire pressure (periodic report) bit1: Tire pressure too high alarm bit2: Tire pressure too low alarm bit3: Tire temperature too high alarm bit4: Sensor abnormal alarm bit5: Tire pressure imbalance alarm bit6: Slow leak alarm bit7: Battery level low alarm
3	Tire Pressure	WORD	Unit: Kpa

5	Tire Temperature	WORD	Unit: °C
7	Battery Level	WORD	Unit: %

Example Data (41+N Bytes):

Scenario: 1 tire, tire pressure too low alarm

Field	HEX Value	Description
Alarm ID	`00 00 00 03`	Sequence 3
Flag Status	`01`	Start flag
Vehicle Speed	`32`	50 km/h
Elevation	`01 F4`	500 meters
Latitude	`0F 42 3D FF`	32,413,583 × 10^{-6}° ≈ 32.413583°
Longitude	`0D ED BA 52`	233,862,360 × 10^{-6}° ≈ 233.862360°
Date Time	`32 04 16 08 1E 2D`	2026-04-22 08:30:45
Vehicle Status	`00 00`	No status bits
Alarm ID Number	`00 00 00 03 00 00 00 03 00 02 03 04 05 06 07 08`	—
Total Alarm/Event List Count	`01`	1 tire
Tire Alarm Position	`01`	Tire position 1
Alarm/Event Type	`00 04`	bit2=1: Tire pressure too low alarm
Tire Pressure	`00 A0`	160 KPa
Tire Temperature	`00 50`	80 °C
Battery Level	`00 64`	100%

Complete HEX:

66 31 03 01 32 01 F4 0F 42 3D FF 0D ED BA 52 32 04 16 08 1E 2D 00 00 00 00 00 03 00 00 00 03 00 02 03 04 05 06 07 08 01 01 00 04 00 A0 00 50 00 64

Total length: 41 + 8 = 49 Bytes

Complete Frame Example (Uplink): 7E 02 00 00 50 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 66 31 03 01 00 00 00 01 00 32 00 64 01 EE 97 8F 07 14 7D 01 E0 26 05 13 14 36 56 00 00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08 01 01 00 00 00 A0 00 50 00 64 D6 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 11 00 01 02 00 00 1C 7E

6 Blind Spot Monitoring Alarm Information (0x67)

Table 6-1 Blind Spot Monitoring Function Alarm Definition Data Format

Offset	Field	Data Type	Description and Requirements
0	Alarm ID	DWORD	Cumulative increment from 0 in alarm order, regardless of alarm type
4	Flag Status	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag

5	Alarm/Event Type	BYTE	**Standard Definitions**: 0x01: Rear approach warning 0x02: Left rear approach warning 0x03: Right rear approach warning **[Manufacturer Custom] Definitions**: 0x04: Front approach warning (IBAMSA) 0xE0: Camera occlusion
6	Vehicle Speed	BYTE	Unit: km/h, range 0~250
7	Elevation	WORD	Altitude in meters (m)
9	Latitude	DWORD	Latitude value in degrees multiplied by 10 to the 6th power
13	Longitude	DWORD	Longitude value in degrees multiplied by 10 to the 6th power
17	Date Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)
23	Vehicle Status	WORD	See Table 6-2
25	Alarm ID Number	BYTE[16]	See Appendix A.4

Table 6-2 Vehicle Status Bit Definitions

Bit	Definition	Description
0	ACC Status	0: Off, 1: On
1	Left Turn Signal Status	0: Off, 1: On
2	Right Turn Signal Status	0: Off, 1: On
3	Wiper Status	0: Off, 1: On
4	Brake Status	0: Not braking, 1: Braking
5	Card Insertion Status	0: Card not inserted, 1: Card inserted
6~9	Reserved	—
10	Positioning Status	0: Not positioned, 1: Positioned
11~15	Reserved	—

Example Data (41 Bytes):

Scenario: Rear approach warning (0x01), start flag

Field	HEX Value	Description
Alarm ID	`00 00 00 04`	Sequence 4
Flag Status	`01`	Start flag
Alarm/Event Type	`01`	Rear approach warning
Vehicle Speed	`32`	50 km/h
Elevation	`01 F4`	500 meters
Latitude	`0F 42 3D FF`	32,413,583 × 10^{-6}° ≈ 32.413583°
Longitude	`0D ED BA 52`	233,862,360 × 10^{-6}° ≈ 233.862360°
Date Time	`32 04 16 08 1E 2D`	2026-04-22 08:30:45

Vehicle Status	`00 00`	No status bits
Alarm ID Number	`00 00 00 04 00 00 00 04 00 02 03 04 05 06 07 08`	—

Complete HEX:

67 29 04 01 01 32 01 F4 0F 42 3D FF 0D ED BA 52 32 04 16 08 1E 2D 00 00 00 00 00 04 00 00 00 04 00 02 03 04 05 06 07 08

Complete Frame Example (Uplink): 7E 02 00 00 47 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 67 29 00 00 00 01 00 01 32 00 64 01 EE 97 8F 07 14 7D 01 E0 26 05 13 14 36 56 00 00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08 1A 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 12 00 01 02 00 00 1F 7E

7 Aggressive Driving Alarm Information (0x70)

Table 7-1 Aggressive Driving Alarm Definition Data Format

Offset	Field	Data Type	Description and Requirements
0	Alarm ID	DWORD	Cumulative increment from 0 in alarm order, regardless of alarm type
4	Flag Status	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag

5	Alarm/Event Type	BYTE	**Standard Definitions**: 0x01: Harsh acceleration warning 0x02: Harsh braking warning 0x03: Harsh cornering warning 0x04: Idle warning 0x05: Abnormal shutdown warning 0x06: Neutral coasting warning 0x07: Engine overspeed warning
6	Alarm Time Threshold	WORD	Unit: seconds
8	Alarm Threshold 1	WORD	When alarm type is 0x01~0x03: Alarm acceleration threshold, unit: 1/100g When alarm type is 0x04~0x07: Alarm vehicle speed threshold, unit: km/h
10	Alarm Threshold 2	WORD	When alarm type is 0x01~0x03: Reserved When alarm type is 0x04~0x07: Alarm engine speed threshold, unit: RPM

12	Vehicle Speed	BYTE	Unit: km/h, range 0~250
13	Elevation	WORD	Altitude in meters (m)
15	Latitude	DWORD	Latitude value in degrees multiplied by 10 to the 6th power
19	Longitude	DWORD	Longitude value in degrees multiplied by 10 to the 6th power
23	Date Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)
29	Vehicle Status	WORD	See Table 7-2
31	Alarm ID Number	BYTE[16]	See Appendix A.4

Table 7-2 Vehicle Status Bit Definitions

Bit	Definition	Description
0	ACC Status	0: Off, 1: On
1	Left Turn Signal Status	0: Off, 1: On
2	Right Turn Signal Status	0: Off, 1: On
3	Wiper Status	0: Off, 1: On
4	Brake Status	0: Not braking, 1: Braking
5	Card Insertion Status	0: Card not inserted, 1: Card inserted
6~9	Reserved	—
10	Positioning Status	0: Not positioned, 1: Positioned

11~15	Reserved	—

Example Data (47 Bytes):

Scenario: Harsh acceleration warning (0x01), start flag, time threshold 2 seconds, acceleration threshold 1.5g

Field	HEX Value	Description
Alarm ID	`00 00 00 05`	Sequence 5
Flag Status	`01`	Start flag
Alarm/Event Type	`01`	Harsh acceleration warning
Alarm Time Threshold	`00 02`	2 seconds
Alarm Threshold 1	`00 96`	150 × 1/100g = 1.5g
Alarm Threshold 2	`00 00`	Reserved
Vehicle Speed	`32`	50 km/h
Elevation	`01 F4`	500 meters
Latitude	`0F 42 3D FF`	32,413,583 × 10^{-6}° ≈ 32.413583°
Longitude	`0D ED BA 52`	233,862,360 × 10^{-6}° ≈ 233.862360°
Date Time	`32 04 16 08 1E 2D`	2026-04-22 08:30:45
Vehicle Status	`00 00`	No status bits
Alarm ID Number	`00 00 00 05 00 00 00 05 00 02 03 04 05 06 07 08`	—

Complete HEX:

70 2F 05 01 01 00 02 00 96 00 00 32 01 F4 0F 42 3D FF 0D ED BA 52 32 04 16 08 1E 2D 00 00 00 00 00 05 00 00 00 05 00 02 03 04 05 06 07 08

Complete Frame Example (Uplink): 7E 02 00 00 4D 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 70 2F 00 00 00 01 00 01 00 02 00 96 00 00 32 00 64 01 EE 97 8F 07 14 7D

01 E0 26 05 13 14 36 56 00 00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08 8B 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 01 00 01 02 00 00 0C 7E

8 IPC Alarm Events (0xA1)

Additional Information ID: 0xA1

Additional Information Length: 72 Bytes

Table 8-1 IPC Alarm Event Data Format

Offset	Field	Length	Description and Requirements
0	Alarm ID	DWORD	Cumulative increment from 0 in alarm order, regardless of alarm type
4	Flag Status	BYTE	0x00: Unavailable 0x01: Start flag 0x02: End flag

5	Alarm/Event Type	BYTE	0x01: Motion detection 0x02: Area alert 0x03: Sound anomaly 0x04: Human detection 0x05: Face detection 0x06: Vehicle detection 0x07: Face recognition 0x08: Collision 0x09: Alcohol test normal 0x0A: Alcohol test incomplete 0x0B: Drinking driving 0x0C: Drunk driving 0x0D: Refuse alcohol test 0xE0: Fire and smoke detection
6	Latitude	DWORD	—
10	Longitude	DWORD	—
14	Date Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)
20	Alarm ID Number	BYTE[16]	Alarm ID number definition, see Appendix A.4

36	Night Vision Mode	BYTE	0x01: Day 0x02: B&W night vision 0x03: Low light full color 0x04: Smart full color
37	Reserved Field	63 bytes	Extended definition based on alarm type

Reserved Field Extension Definition (when alarm type = 0x08~0x0C):

Offset	Field	Description
0	Extension Type	0x08: License plate content 0x09~0x0C: Alcohol concentration
—	License Plate Content	When extension type = 0x08, represents license plate content
—	Alcohol Concentration	When extension type = 0x09~0x0C: byte[0]: Alcohol test mode (1: Active; 2: Passive; 4: Refuse test) byte[1]: Alcohol concentration string length, max 30 byte[2-31]: ASCII string value + unit

Example Data (72 Bytes):

Scenario: Human detection (0x04), day night vision mode

Field	HEX Value	Description

Alarm ID	`00 00 00 06`	Sequence 6
Flag Status	`01`	Start flag
Alarm/Event Type	`04`	Human detection
Latitude	`0F 42 3D FF`	32,413,583 × 10^{-6}° ≈ 32.413583°
Longitude	`0D ED BA 52`	233,862,360 × 10^{-6}° ≈ 233.862360°
Date Time	`32 04 16 08 1E 2D`	2026-04-22 08:30:45
Alarm ID Number	`00 00 00 06 00 00 00 06 00 02 03 04 05 06 07 08`	—
Night Vision Mode	`01`	Day
Reserved Field	63 × `00`	Reserved extension

Complete HEX:

A1 48 06 01 04 0F 42 3D FF 0D ED BA 52 32 04 16 08 1E 2D 00 00 00 06 00 00 00 06 00 02 03 04 05 06 07 08 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Complete Frame Example (Uplink): 7E 02 00 00 66 12 34 56 78 90 12 00 01 00 00 00 00 00 00 00 01 01 A1 48 00 00 00 01 00 0C 01 EE 97 8F 07 14 7D 01 E0 26 05 13 14 36 56 00 00 00 01 00 00 00 01 00 02 03 04 05 06 07 08 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 C0 7E

Complete Frame Example (Downlink): 7E 80 01 00 05 12 34 56 78 90 12 00 00 00 01 02 00 00 0D 7E

Appendix A Alarm Attachment Upload Process

A.1 Process Overview

Alarm attachment upload is the core business process in the Active Safety Intelligent Prevention and Control System, used to upload alarm scene attachments (pictures, audio, video, text, etc.) generated by the terminal to the platform attachment server.

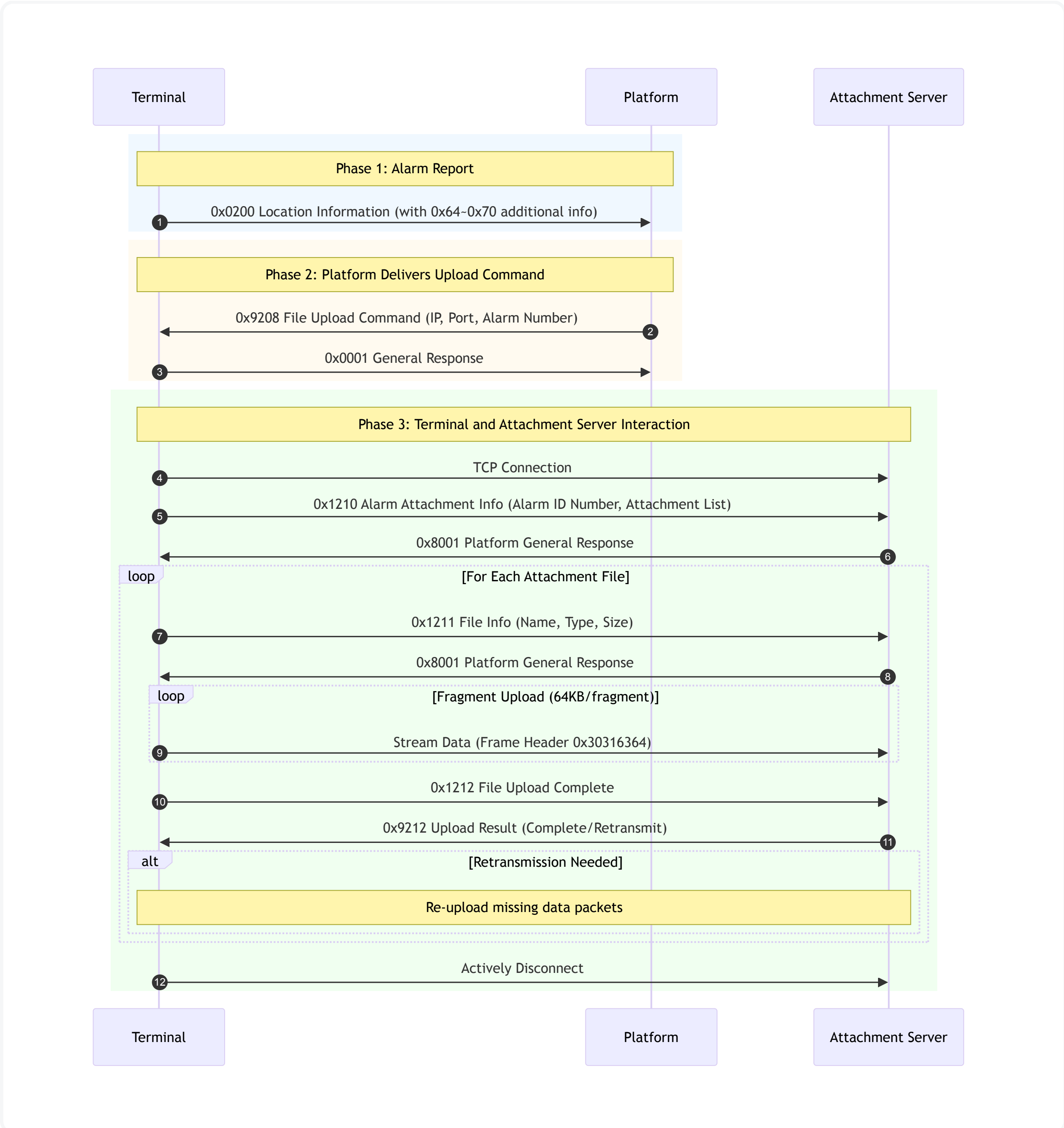
This process involves three participants: **Terminal**, **Platform (JT808 Server)**, and **Attachment Server**.

A.2 Process Phases

Phase	Participant	Message	Description
1. Alarm Report	Terminal → Platform	0x0200 (with additional info 0x64~0x70)	After detecting an alarm, the terminal reports location information with alarm info
2. Command Delivery	Platform → Terminal	0x9208	After receiving alarm with attachment, platform delivers attachment upload command
3. Terminal Response	Terminal → Platform	0x0001 (General Response)	Terminal responds after receiving command
4. Server Connection	Terminal → Attachment Server	TCP Connection	Terminal connects to attachment server based on IP and port in command
5. Attachment Info Report	Terminal → Attachment Server	0x1210	Report alarm number, attachment count and list
6. File Info Report	Terminal → Attachment Server	0x1211	Report file name, type, size for each file

7. File Data Upload	Terminal → Attachment Server	Stream data (0x30316364)	Fragment upload file data, default 64KB/fragment
8. File Complete Report	Terminal → Attachment Server	0x1212	Sent after single file transmission is complete
9. Server Response	Attachment Server → Terminal	0x9212	If retransmission needed, terminal re-uploads missing packets
10. Disconnect	Terminal → Attachment Server	—	Terminal actively disconnects after all files are sent

A.3 Sequence Diagram



A.3.1 Vehicle Status Data Recording File

The vehicle status data recording file is a binary file that records vehicle status data in continuous data blocks. The data block format is as follows:

Offset	Field	Data Type	Description and Requirements
0	Total Data Block Count	DWORD	Total number of data blocks in the recording file

4	Current Data Block Number	DWORD	Current data block number in the recording file
8	Alarm Flag	DWORD	Reference JT808 definition
12	Vehicle Status	DWORD	Reference JT808 definition
16	Latitude	DWORD	Latitude value in degrees multiplied by 10 to the 6th power, accurate to 1/1,000,000 degree
20	Longitude	DWORD	Longitude value in degrees multiplied by 10 to the 6th power, accurate to 1/1,000,000 degree
24	Satellite Elevation	WORD	Satellite altitude in meters (m)
26	Satellite Speed	WORD	1/10 km/h
28	Satellite Direction	WORD	0-359, North is 0, clockwise
30	Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)
36	X-axis Acceleration	WORD	In g multiplied by 10 to the 2nd power, accurate to 1/100 g
38	Y-axis Acceleration	WORD	In g multiplied by 10 to the 2nd power, accurate to 1/100 g

40	Z-axis Acceleration	WORD	In g multiplied by 10 to the 2nd power, accurate to 1/100 g
42	X-axis Angular Velocity	WORD	In degrees per second multiplied by 10 to the 2nd power, accurate to 1/100 degree per second
44	Y-axis Angular Velocity	WORD	In degrees per second multiplied by 10 to the 2nd power, accurate to 1/100 degree per second
46	Z-axis Angular Velocity	WORD	In degrees per second multiplied by 10 to the 2nd power, accurate to 1/100 degree per second
48	Pulse Speed	WORD	1/10 km/h
50	OBD Speed	WORD	1/10 km/h
52	Gear Status	BYTE	0: Neutral 1-9: Gear 10: Reverse 11: Park
53	Accelerator Pedal Position	BYTE	Range 1-100, unit: %
54	Brake Pedal Position	BYTE	Range 1-100, unit: %
55	Brake Status	BYTE	0: No brake 1: Brake

56	Engine Speed	WORD	Unit: RPM
58	Steering Wheel Angle	WORD	Angle turned by steering wheel, clockwise is positive, counterclockwise is negative
60	Turn Signal Status	BYTE	0: No turn signal 1: Left turn signal 2: Right turn signal
61	Reserved	BYTE[2]	—
63	Check Bit	BYTE	Accumulated sum from first byte to byte before check bit, then take the low 8 bits as check code

A.4 Alarm ID Number Format

The alarm ID number is the unique identifier for an alarm event, used to associate an alarm with its attachments. The alarm ID number format is as follows:

Offset	Field	Data Type	Description and Requirements
0	Terminal ID	BYTE[7]	7 bytes, composed of uppercase letters and numbers
7	Time	BCD[6]	YY-MM-DD-hh-mm-ss (GMT+8 time)

13	Sequence Number	BYTE	Sequence number for alarms at the same time point, cumulative increment from 0
14	Attachment Count	BYTE	Number of attachments associated with this alarm
15	Reserved	BYTE	—

A.5 Message Definition Summary

Message ID	Name	Direction	Message Type	Description
0x0200	Location Information Report	Terminal→Platform	Signaling Data	With alarm additional info (0x64~0x70)
0x9208	File Upload Command	Platform→Terminal	Signaling Data	Deliver attachment server address and alarm number
0x1210	Alarm Attachment Info	Terminal→Attachment Server	Signaling Data	Report attachment list
0x1211	File Info Upload	Terminal→Attachment Server	Signaling Data	Report single file info
0x1212	File Upload Complete	Terminal→Attachment Server	Signaling Data	Single file transmission complete
0x9212	File Upload Complete Response	Attachment Server→Terminal	Signaling Data	Upload result/retransmission list

0x0001	General Response	Bidirectional	Signaling Data	General acknowledgment response

A.6 Attachment Server Address Information (0x9208)

Offset	Field	Data Type	Description and Requirements
0	Attachment Server IP Address Length	BYTE	Length k
1	Attachment Server IP Address	STRING	Server IP address
1+k	Attachment Server Port (TCP)	WORD	Server port number when using TCP transmission
3+k	Attachment Server Port (UDP)	WORD	Server port number when using UDP transmission
5+k	Alarm ID Number	BYTE[16]	Alarm ID number definition
21+k	Alarm Number	BYTE[32]	Unique number assigned by platform for the alarm
53+k	Reserved	BYTE[16]	—

A.7 Alarm Attachment Information (0x1210)

Offset	Field	Data Type	Description and Requirements

0	Terminal ID	BYTE[7]	Composed of uppercase letters and numbers, padded with 0x00 if less than 7 bytes
7	Alarm ID Number	BYTE[16]	Alarm ID number
23	Alarm Number	BYTE[32]	Unique alarm number assigned by platform
55	Info Type	BYTE	0x00: Normal alarm file info 0x01: Retransmission alarm file info
56	Attachment Count	BYTE	Number of attachments associated with the alarm
57	Attachment Info List	—	See Appendix A.7.1 Alarm Attachment Message Data Format

Note: When the connection between the terminal and the attachment server is abnormally disconnected during the alarm attachment upload process, after the connection is restored, the terminal needs to resend the alarm attachment information message. The attachment files in the message should be those that were not uploaded and not completed before the disconnection.

A.7.1 Alarm Attachment Message Data Format

Offset	Field	Data Type	Description and Requirements
0	File Name Length	BYTE	Length k
1	File Name	STRING	File name string

1+k	File Size	DWORD	Size of the current file

A.8 File Info Upload (0x1211)

Offset	Field	Data Type	Description and Requirements
0	File Name Length	BYTE	File name length l
1	File Name	STRING	File name
1+l	File Type	BYTE	0x00: Picture 0x01: Audio 0x02: Video 0x03: Text 0x04: Other
2+l	File Size	DWORD	Size of the current uploaded file

A.9 File Data Upload

Message Type: Stream Data Message

After the terminal sends the file info upload command and receives a response, it sends file data to the attachment server. The payload package format is defined as follows:

Offset	Field	Data Type	Description and Requirements
0	Frame Header	DWORD	Fixed to 0x30 0x31 0x63 0x64
4	File Name	BYTE[50]	File name
54	Data Offset	DWORD	Current transmission file data offset
58	Data Length	DWORD	Payload data length

62	Data Body	BYTE[n]	Default length 64K, if file is less than 64K then actual length

Note: When the attachment server receives the file stream reported by the terminal, it does not need to respond.

A.10 File Upload Complete (0x1212)

Offset	Field	Data Type	Description and Requirements
0	File Name Length	BYTE	l
1	File Name	STRING	File name
1+l	File Type	BYTE	0x00: Picture 0x01: Audio 0x02: Video 0x03: Text 0x04: Other
2+l	File Size	DWORD	Size of the current uploaded file

A.11 File Upload Complete Response (0x9212)

Offset	Field	Data Type	Description and Requirements
0	File Name Length	BYTE	l
1	File Name	STRING	File name
1+l	File Type	BYTE	0x00: Picture 0x01: Audio 0x02: Video 0x03: Text 0x04: Other

2+I	Upload Result	BYTE	0x00: Complete 0x01: Retransmission needed
3+I	Retransmission Packet Count	BYTE	Number of packets needing retransmission, 0 if none
4+I	Retransmission Packet List	—	See Appendix A.12 Retransmission Packet Information

A.12 Retransmission Packet Information

Offset	Field	Data Type	Description and Requirements
0	Data Offset	DWORD	Offset of data needing retransmission in the file
4	Data Length	DWORD	Length of data needing retransmission

A.13 File Naming Rules

Attachment file name naming format:

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Field	Description
File Type	00: Picture; 01: Audio; 02: Audio/Video; 03: Text; 04: Other

Channel Number	0~37: JT/T 1076 video channel; 64: Driving assistance module; 65: Driver behavior monitoring module; 0: Not related to channel
Alarm Type	Composed of peripheral ID and module alarm type, e.g., forward collision warning is "6401"
Sequence Number	Sequence number to distinguish files of the same channel and type
Alarm Number	Unique number assigned by platform for the alarm
Extension	jpg/png/wav/h264/mp4/bin

File Name Example:

00_64_6401_001_20260422001.jpg

Means: Picture_Driving assistance channel_Forward collision alarm_Sequence 1_Alarm number 20260422001